The Islamia University of Bahawalpur
Department of Information Technology

INFT-4041 · BSIT · 5th Semester · Fall 2026

E-Commerce Application Development

Appendix

F

The Boundary With Other Courses

Taught in	
Laboratory	
Assessment	

This is a class extract. It is *Appendix F* of the unified handout, printed on its own so that a week's reading is a week's reading. Page numbers restart at 1 and are local to this extract; chapter, figure and table numbers are the ones the full volume gives them, so a reference you follow here lands in the same place there.

Cross-references to other chapters print as their numbers in the full volume. For the whole course in one file — front matter, the HEC tracing table, all twenty-three chapters and all seven appendices — see `ECAD_Fall2026_Handout.pdf`.

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Chapter F

The Boundary With Other Courses

This course sits downstream of four others and alongside a fifth. It assumes a good deal and repeats a little, and both the assumptions and the repetitions are deliberate. This appendix records what belongs where, so that a lecturer can see what to assume and a student can see why a familiar topic reappears.

F.1 The Edition History, Which Is Unusual

Before the boundaries, the provenance — because this course's position in the degree is less settled than any of its siblings'.

Edition	Title	Where	Content published?
2017	E-Commerce	BSSE elective, p. 38	**Yes — p. 94, the only full specification in any edition**
2023	—	**Absent entirely**	One incidental mention, p. 23
2025	Digital Marketing & E-Commerce	Interdisciplinary / allied list, item 11	No — title and credit hours only

So the single published outline for this course is a **2017 specification for an elective in a different degree**, the current editions do not specify it at all, and the department teaches it as **INFT-4041, BSIT, 5th semester, 3 credit hours**. Everything in this handout is traced to that 2017 outline, with each addition justified in the Course Specification at the front.

A prerequisite under another name

HEC's 2017 entry gives the prerequisite as **Web Engineering**. The department has no course of that name; its equivalent is **CS-207 / INFT-4005 Web Technologies** at 4th semester.

The name differs and the dependency is satisfied. Recorded here rather than silently matched, because a reviewer comparing the two documents will notice, and the answer should be written down before they ask.

F.2 Web Technologies — the Prerequisite

The rule

Web Technologies owns the request. This course owns what is at the other end of it.

A student arriving here should be able to read an HTTP request and response, write HTML and CSS, and know what a form submission is. What this course adds is *an application that answers*, with a database behind it, money passing through it, and consequences attached to getting it wrong.

Topic	Owned by	Here
HTTP methods, status codes, headers	Web Technologies	Assumed; used in Chapter 16 and Chapter 21
HTML, CSS, responsive layout	Web Technologies	Assumed; Chapter 8 is about *overriding* templates, not writing CSS
Cookies, as a browser feature	Web Technologies	Chapter 9 asks where the state they point at actually lives
Forms and validation	Web Technologies	Chapter 10 is about the *sequence* of forms and what each commits
A server-side language	Web Technologies	Appendix B is a refresher, not a first course

F.3 Database Systems

The rule

Database Systems owns the model. This course owns the consequences of somebody else's model.

You will not design a schema here. You will read one you did not write, find out what it costs, and discover that the answer changes with its size.

- **Assumed:** SQL selects and joins, primary and foreign keys, normalisation, what an index is.
- **Added here:** reading the query a screen *actually* ran (Chapter 6); measuring an index rather than assuming it (Chapter 6 found two that made the page slower at 50,000 products); and a schema whose shape is a historical record of somebody's decisions — 18 product tables, options that grow by addition and variants that grow by multiplication (Chapter 7).
- **The useful repetition:** every Database Systems student has been told that indexes make queries faster. Chapter 6 is where that becomes a measurement with a crossover point in it.

F.4 Information Security

The rule

Information Security owns the principles. Chapter 16 owns one shop, measured twice.

That chapter is not a survey of attacks. It is a battery of fifteen checks run against a real installation before and after one configuration file, and its central finding — that the famous attacks were already handled and every real problem was in the deployment — is one a principles course cannot deliver.

- **Assumed:** the CIA triad, authentication versus authorisation, hashing versus encryption, what injection and cross-site scripting *are*.

- **Added here:** measuring them (Chapter 16); the cost factor as a trade-off with a number attached (Chapter 15); a timing side channel caused by the defence itself; and **the legal boundary**, which is stated in Appendix A, in Chapter 16 and again in Chapter 22.
- **Not covered here:** cryptographic construction, network security, forensics. Those stay with Information Security.

F.5 Software Engineering

- **Assumed:** requirements, version control, testing, the idea that a system has a lifecycle.
- **Added here:** the build-versus-buy-versus-extend decision made against a real platform (Chapter 4); extension as the discipline that keeps a system upgradable (Chapter 19); and deployment measured rather than described (Chapter 20).
- **The overlap worth naming:** Chapter 4's decision is a software engineering decision, taught here because it is the one that determines everything after it in this course. A student who has made it in Software Engineering should make it again here with a specific platform in front of them, which is a different exercise.

F.6 Virtual Systems and Services — the Same Semester

⚠ This overlap needs stating, because the two courses are taught side by side

INFT-4008 Virtual Systems and Services is also a 5th-semester BSIT course, and it also uses containers. A student taking both will meet Docker twice in one semester.

The split. Virtual Systems and Services owns *virtualisation as a subject*: hypervisors, container internals, orchestration, cloud service models, how a container differs from a virtual machine at the kernel level.

This course uses containers as a tool and says so. Chapter 5 spends one chapter on what a container is — enough to use Compose confidently and no more — because its purpose is a reproducible shop, not an education in virtualisation. Chapter 20 returns to deployment, again from the application's side: saturation, backup, restore.

For the lecturer: if the two courses are taught in the same semester, Chapter 5 can be shortened by agreement, and the time spent on Chapter 6, which always needs more. If they are not, Chapter 5 stands alone as written.

F.7 What This Course Does Not Teach

Stated so that nobody is surprised at the end of it.

- **Front-end frameworks.** No React, no Vue. Chapter 8 overrides templates in the platform's own engine.
- **Writing a cart from scratch.** Chapter 4 argues explicitly against it, and the argument is the lesson.
- **Digital marketing as a discipline.** Chapter 21 covers what is checkable and says plainly what it is leaving out, which matters because the 2025 edition renames the course.
- **Cloud deployment on a named provider.** Chapter 20 measures a deployment; which cloud it sits on is a different course's subject and dates faster than this handout.
- **Payment integration with a live provider.** Chapter 14 ships its own, because a merchant account cannot be a prerequisite for a laboratory.